MIL-R-51081A(MU)
20 January 1971
SUPERSEDING
MIL-R-51081(CmlC)
29 March 1962

MILITARY SPECIFICATION

RADIOACTIVE SOURCE, COBALT 60, GAMMA, M1A1

1. SCOPE

1.1 This specification covers a capsulated cobalt 60 source.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-6866 - Inspection, Penetrant Method of.
MIL-M-19590 - Marking of Commodities and Containers to Indicate Radioactive Material.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

LM 124-10-10 - Radioactive Source, Cobalt 60, Gamma, M1A1.
D150-2-1 - Marking Diagram for Shipping Containers.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6665

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, IL 60606.)

CODE OF FEDERAL REGULATIONS

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Officer, Washington, D.C. 20402. Orders for the above publication should cite "49 CFR 171-179.")

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM STANDARD

D999-68 - Vibration Test for Shipping Containers.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing LM 124-10-10 or on the subsidiary drawings shall conform to the specification thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the source shall conform to the specifications and drawings listed on Drawing LM 124-10-10 and subsidiary drawings.

3.2 Manufacture and assembly. The radioactive source shall be manufactured and assembled in accordance with C124-10-10.

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3.3 Performance.

3.3.1 Activity. The cobalt 60 radioactive source shall have an activity between 103.5 millicuries and 126.5 millicuries (see 3.3.1.1) when tested in accordance with 4.4.4.1.

3.3.1.1 Calibration. The strength (milliroentgens per hour at 1 meter) and the activity (3.3.1) on date of calibration shall have an accuracy of plus or minus 3 percent when tested in accordance with 4.4.4.1. Compliance with this requirement shall be determined while testing for activity (3.3.1).

3.3.1.2 Calibration curve. A calibration curve conforming to figure I shall be furnished with each source. The curve shall be constructed using semi-log graph paper 1 cycle x 12 divisions per inch. The curve and the printing shall be produced using black indelible ink.

3.3.2 Radioactive leakage. The radioactive source shall show no leakage or transfer of radioactive material greater than 5×10^{-4} microcuries after rough handling for 2 hours at a force of at least one times gravity as specified in 4.4.4.2. The source shall be set aside for seven days and retested only for removable activity.

3.3.3 Special form material requirements. The radioactive source shall show no evidence of radioactive leakage (3.3.2), cracks or fissures, and shall not melt or ignite, or undergo a weight change greater than 0.005 percent when subjected to the conditions and tested in accordance with table I (see 4.3.2.1) and 4.3.3.

3.4 Preproduction. Prior to the start of regular production, preproduction samples of radioactive sources and inert sources shall be produced in accordance with this specification for examination and tests (see 4.3).

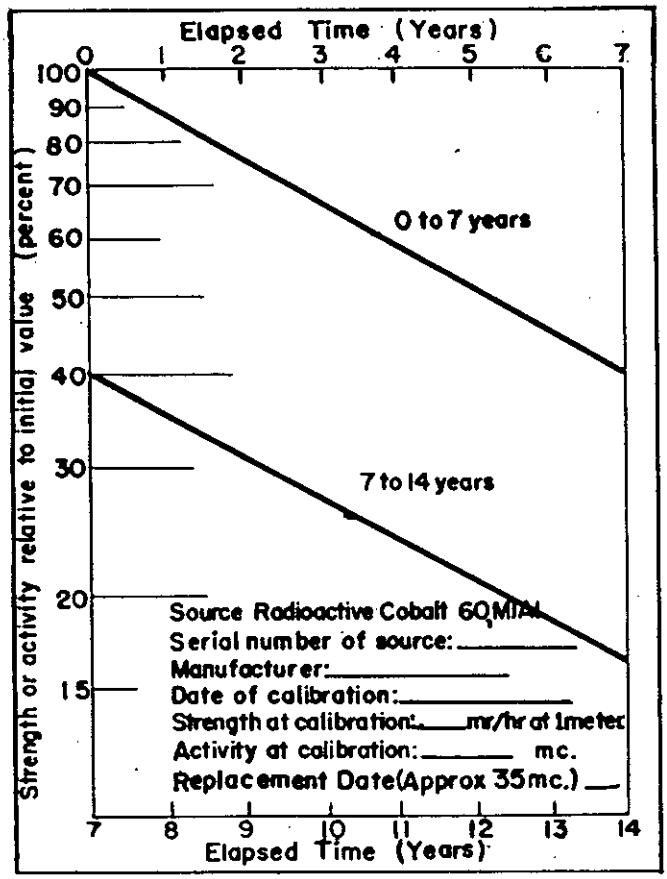
3.5 Workmanship. The radioactive sources shall be free from cracks, dents, broken chain links or ring, and foreign matter such as grease, oil, or viscous material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

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NOTES:

1. Scale: approximately one-half
2. The 0-to-7-years portion of the curve shall be a straight line ruled through the points (0 years 100 percent) and (7 years 40 percent).
3. The 7-to-14-years portion of the curve shall be a straight line ruled through the points (7 years, 40 percent) and (14 years, 16 percent).
4. The blank spaces shown in the figure shall be filled with the specified data.

Figure 1 — Calibration curve

PART OF SPEC	CHEMICAL CORPS SPECIFICATION FIGURE	
MIL-S-51081A	DATE: 20 JAN 71	A18-60-446

Figure 1. (Drawing A18-60-446)

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4.1.2 Government's responsibility. The Government will be responsible for performance of the tests in 4.3.3. Samples shall be forwarded to the laboratory designated by the contracting officer.

4.1.3 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of five cobalt 60 radioactive sources and five inert capsules (Dwg C124-10-34) shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination and nondestructive tests. Each radioactive source in the preproduction sample shall be examined for the defects listed in the list of defects (4.4.3.2) and tested in accordance with 4.4.4; if any source is found defective the preproduction lot shall be rejected. The five inert capsules and five radioactive sources shall be tested in accordance with table I (see 4.1.2); any radioactive source or inert capsule found defective shall reject the preproduction sample.

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Table I - Special form material requirements*

Condition	5 Inert	5 Radioactive	Requirement	Test
Free drop	X	X	3.3.3	4.3.3.1
Leakage		X	3.3.2	4.4.4.2
Dye penetrant	X		3.3.3	4.3.3.6
Percussion	X	X	3.3.3	4.3.3.2
Leakage		X	3.3.2	4.4.4.2
Dye penetrant	X		3.3.3	4.3.3.6
Heating	X	X	3.3.3	4.3.3.3
Leakage		X	3.3.2	4.4.4.2
Dye penetrant	X		3.3.3	4.3.3.6
Immersion	X	X	3.3.3	4.3.3.4
Leakage		X	3.3.2	4.4.4.2
Dye penetrant	X		3.3.3	4.3.3.6
Vibration	X	X	3.3.3	4.3.3.4
Leakage		X	3.3.2	4.4.4.2
Dye penetrant	X		3.3.3	4.3.3.6
Thermal shock	X	X	3.3.3	4.3.3.5
Leakage		X	3.3.2	4.4.4.2
Dye penetrant	X		3.3.3	4.3.3.6

*Note: The samples shall be tested and examined for compliance with 3.3.2 and the applicable part of 3.3.3 in the sequence shown in the table.

4.3.3 Tests. Chains shall be removed from the sources prior to performing these tests. Tests shall be performed as follows:

4.3.3.1 Free drop. Drop the sources through a distance of 30 feet onto a flat unyielding horizontal surface so that the source strikes the surface in such a position as to suffer maximum damage.

4.3.3.2 Percussion. Place the source on a sheet of lead which is not more than 1 inch thick, has a hardness number between 3.5 to 4.5 on the VICKERS scale, and is supported by a smooth unyielding surface. Drop a flat circular end of a 1 inch in diameter steel rod weighing 3 pounds through a distance of 40 inches so that the face of the flat circular end impacts on the source.

4.3.3.3 Heating. Equilibrate the source to a temperature of 1475°F in a muffle furnace and maintain this temperature for a period of 10 minutes.

4.3.3.4 Immersion. Immerse the source for 24 hours in water at room temperature. The water shall be PH6 to PH8 with a maximum conductivity of 10 micromhos per centimeter. Weight before and after test on a balance accurate to 0.001 gram. Record weight change.

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4.3.3.5 Thermal shock. Subject the source to a temperature of -40°C for 2 hours. Remove source from the cold environment and immediately subject it to a temperature of +50°C ± 2°C for 2 hours. This procedure shall constitute one complete cycle of the thermal test. Test duration shall be three cycles.

4.3.3.6 Dye penetrant. Apply dye to inert capsules in accordance with MIL-I-6866. Examine the capsules for evidence of any cracks or fissures.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the radioactive sources produced by one manufacturer, at one plant, from the same materials, and under the same manufacturing conditions.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Each source shall be examined and tested in accordance with 4.4.3.

4.4.3 Inspection procedures.

4.4.3.1 For examination and nondestructive tests. Each source shall be examined in accordance with the list of defects (4.4.3.2) and tested in accordance with 4.4.4.

4.4.3.2 List of defects.

(a) Radioactive source, cobalt 60, gamma, M1A1 (Drawing C124-10-10).

1. Marking missing, incorrect, or illegible.
2. Components missing, incorrect, or incorrectly assembled.
3. Calibration curve missing, incorrect, or incomplete (see figure 1).
4. Workmanship (see 3.5).

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Activity. Determine the calibrated activity and strength of the source with a standard radiation measuring instrument which has been previously calibrated with a cobalt 60 source certified by National Bureau of Standards (see 6.3 and 6.4).

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4.4.4.2 Radioactive leakage. Subject the source to a vibration test in accordance with ASTM Method D999-68. However, for this test replace the shipping container by a metal screw-cap container of appropriate size. Place the source within the container and place the container on the specified table. Vibrate the table for 2 hours at a speed that will yield output forces acting on the container equal to at least one times gravity. Remove the source from the container after vibration is complete and wipe all external surfaces with high wet-strength filter paper moistened with distilled water. Allow the filter paper to dry and then determine any contamination present by using a scaler equipped with a scintillation or Geiger-Mueller tubes for counting. Counting equipment shall be capable of detecting 1×10^{-4} microcuries of radioactive material on the filter paper. Repeat the wipe test on the source after seven days elasped time.

4.4.5 Acceptance/rejection criteria. If any source fails when examined in accordance with 4.4.3 and tested in accordance with 4.4.4 it shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level C. The radioactive source shall be packaged to afford adequate protection against deterioration and damage in shipment from the supply source to the first receiving activity for immediate use. Packaging shall be in compliance with applicable rules and regulations of Department of Transportation (DOT) and Atomic Energy Commission (AEC).

5.2 Packing, level C. The radioactive sources, packaged as specified in 5.1 shall be packed in accordance with applicable requirements specified in DOT regulation 49 CFR 171-179 to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall comply with Uniform Freight Classification rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages shall be marked in compliance with DOT regulation 49 CFR 171-179, AEC, MIL-M-19590 and applicable requirements of Drawing D150-2-1.

6. NOTES

6.1 Intended use. The source ocvered by this specification is intended to provide a source of gamma radiation for use in the M3A1.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or no to proceed with production.

6.3 Instruments. The following instruments have been found to be satisfactory for determining the activity:

Victoreen R Chamber
Landeverik R Meter

6.4 Caution. Care should be exercised to prevent injury to personnel engaged in handling and testing radioactive sources. The National Bureau of Standards handbook (NSB No. 73) and Atomic Energy Commission Regulation 10 CFR 20-40 contain information pertaining to protection of personnel from radioactive emanations.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 6665-A282

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OFFICIAL BUSINESS

Commanding Officer
Edgewood Arsenal
ATTN: SMUEA–QAES
Edgewood Arsenal, Maryland 21010

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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